Year 9 Science – Booklet 5: Physics

Waves & EM Spectrum — ANSWERS

Note: despite the booklet's cover title, every page actually included covers Sound (waves, speed, loudness/pitch, the ear, echoes/ultrasound) – there is no EM spectrum content in the pages provided, so none is answered below.

2.1 Waves

In-text questions

A. Amplitude, frequency and wavelength.

B. Parallel to (in the same direction as) the direction the wave travels.

Spot the word a). Amplitude.

Spot the word b). Compression.

C. The incident wave.

Summary Questions

1. Choose the correct bold word (5 marks)

A wave is an oscillation or vibration that transfers **energy**. The distance from the centre to the top of the wave is the **amplitude**. The distance from one crest to the next crest is the **wavelength**. Waves can **reflect** when they hit a barrier, and cancel out or add up when they **superpose**.

2. Compression vs rarefaction (2 marks)

A compression is where the coils of the spring are pushed close together; a rarefaction is where the coils are spread further apart than normal.

3. Longitudinal vs transverse waves, with examples (6 marks QWC)

In a transverse wave the oscillations are at right angles (90°) to the direction of travel – e.g. light waves, or a slinky/string shaken side to side. In a longitudinal wave the oscillations are parallel to the direction of travel, creating alternating compressions (particles close together) and rarefactions (particles spread apart) – e.g. sound waves, or a slinky pushed and pulled along its length.

2.2 Sound and energy transfer

In-text questions

A. A vibration (a vibrating object).

B. 340 m/s (in air).

Stormy night box

a). distance = speed × time = 340 m/s × 4 s = **1360 m (1.36 km)**.

b). The thunder and lightning would happen at almost the same time, with no noticeable delay, since the storm is right overhead (distance close to zero).

C. Solid, liquid, gas.

Summary Questions

1. Copy and complete (5 marks)

Sound is produced by objects that are **vibrating**. This makes the air molecules **move backwards and forwards** and produces a sound wave. Sound travels fastest in **solids** and slowest in **gases**, and it cannot travel through a **vacuum**.

2. Why sound is slower in a gas than a liquid (2 marks)

In a gas the particles are much further apart than in a liquid, so it takes longer for a vibration to pass from one particle to the next, making the wave travel more slowly than in a liquid, where the particles are closer together.

3. Compare time for light vs sound over the same distance (6 marks QWC)

Light would reach your eye almost instantly (300 000 000 m/s), whereas sound takes a much longer, measurable time (340 m/s in air) – e.g. over 10 m, light takes about 0.000000033 s while sound takes about 0.03 s, so light arrives roughly a million times faster. This is why you see something happen (e.g. a teacher's mouth move) before you hear the sound it produces.

2.3 Loudness and pitch

In-text questions

A. Amplitude.

B. Frequency.

Conversions box

a). 0.02–20 kHz (20 Hz to 20 000 Hz).

b). 1–123 kHz (1000 Hz to 123 000 Hz).

Summary Questions

1. Choose the correct bold word (4 marks)

The loudness of a sound depends on the **amplitude** and the pitch of the sound depends on the **frequency**. Frequency is measured in **hertz**. The range of frequencies you can hear is called the **audible** range.

2. Human vs dolphin hearing range (2 marks)

Humans hear from 20 Hz to 20 000 Hz. Dolphins hear a much wider range, from 1000 Hz up to 100 000 Hz – so dolphins can hear far higher-pitched sounds than humans, though (unlike humans) they can't hear the very lowest-pitched sounds below 1000 Hz.

3. How vocal chords vary pitch and loudness (6 marks QWC)

To make a louder sound, the vocal chords vibrate with a bigger amplitude (more vigorously), producing a larger-amplitude sound wave. To make a higher-pitched sound, they vibrate faster (at a higher frequency), producing more waves per second; a lower pitch comes from vibrating more slowly. Loudness and pitch vary independently, so a note can be any combination of loud/quiet and high/low pitched depending on both the amplitude and frequency of the vibration.

2.4 Detecting sound

In-text questions

A. The eardrum.

B. Exposure to very loud sounds or a head injury can permanently damage hearing; a build-up of ear wax or a hole in the eardrum can also impair hearing (though the eardrum can heal).

Summary Questions

1. Copy and complete (10 marks)

When a sound wave enters your ear it makes the **eardrum** vibrate. This makes the **ossicles** vibrate. The **ossicles** vibrate and this makes the liquid inside your **cochlea** vibrate. Cells at the base of **hairs** inside your **cochlea** produce an **electrical** signal that travels up your **auditory nerve** to your brain. Sound intensity is measured in **decibels**. Your hearing can be **damaged** by loud sounds. In a microphone a **diaphragm** vibrates, which produces an electrical signal.

2. Temporary vs permanent hearing damage (2 marks)

Not permanent: a build-up of ear wax, or a burst eardrum (which can heal/grow back). Permanent: prolonged exposure to very loud noise, or a severe head injury, can permanently damage the hair cells in the cochlea or the auditory nerve, which don't repair themselves.

3. Compare the ear and a microphone (6 marks QWC)

Both detect sound waves using a vibrating part: the eardrum in the ear, the diaphragm in a microphone. In the ear, this vibration is passed on and amplified by the ossicles, causing the liquid and hair cells in the cochlea to produce an electrical signal sent via the auditory nerve to the brain, where it is interpreted as sound. A microphone converts the diaphragm's vibration directly into an electrical signal that can be sent to an amplifier or recorder instead of a brain. So both convert sound waves into an electrical signal carrying the same information, but the ear's signal goes to the brain for interpretation, while the microphone's goes to external equipment.

2.5 Echoes and ultrasound

In-text questions

A. An echo is the reflection of a sound wave off a surface, heard as a repeat of the sound after a short delay.

B. Above 20 000 Hz (20 kHz).

How deep? box

Total distance = speed × time = 1500 m/s × 1.6 s = 2400 m. Depth = 2400 ÷ 2 = **1200 m** (dividing by 2 since the pulse travels down and back).

C. Any one of: prenatal (unborn baby) scanning, sonar (ocean depth/finding fish), physiotherapy (treating tendons), cleaning jewellery, cancer detection.

Summary Questions

1. Copy and complete (9 marks)

An echo is a **reflection** of sound. You can use the **time (delay)** between making a sound and hearing an echo from a surface to calculate the **distance** to it. Soft materials **absorb** sound and reduce echoes. Animals use ultrasound to **communicate** and **navigate**. Ultrasound is used to make an **image** of a fetus, or break down **kidney stones**. Fishermen can use ultrasound to find the **depth** of the ocean.

2. One way a doctor might use ultrasound (2 marks)

To scan a pregnant woman's womb: the machine sends ultrasound waves in, detects the echoes reflecting off the fetus, and uses the return time of the echoes to build up an image.

3. Presentation for fishermen on using sonar (6 marks)

A transmitter under the boat sends a beam of ultrasound down into the water. It travels until it hits an object, such as a shoal of fish or the seabed, and reflects back as an echo. A receiver on the boat detects the echo and measures how long it took to return. Since we know the speed of sound in water (about 1500 m/s), we can calculate the depth using depth = (speed × time) ÷ 2 (dividing by 2 because the pulse travels there and back). Sending pulses continuously builds up a picture of where shoals of fish are and how deep they are beneath the boat.

P2.1 Waves (workbook)

A. Fill in the gaps

In science, a **wave** is an oscillation or vibration that transfers **energy** or information. **Amplitude** is the distance from the middle to the top of a wave. **Frequency** is the number of waves that go past a particular point per second. **Wavelength** is the distance from one point on a wave to the same point on the next wave. The top of a wave is called a **peak** or crest, and the bottom of a wave is called a **trough**. In a **transverse** wave, the oscillation is at 90° to the direction of the wave. In a **longitudinal** wave, the oscillation is parallel to the direction of the wave. In a compression the coils of the spring are **close together**. In a rarefaction the coils are **further apart**. Waves bounce off surfaces and barriers. This is called **reflection**. The wave coming into the barrier is called the **incident** wave and the wave bouncing off the barrier is called the **reflected** wave.

B. a) Label the wave diagram

Vertical distance from the centre line to a peak = **amplitude**; horizontal distance between two equivalent points on consecutive waves = **wavelength**; the top points = **peaks/crests**; the bottom points = **troughs**.

B. b) Type of wave shown

A transverse wave.

C. Similarity/difference: transverse vs longitudinal

Similarity: both transfer energy without transferring matter. Difference: in a transverse wave the oscillations are at right angles to the direction of travel; in a longitudinal wave they are parallel to it.

D. a) What happens when a wave hits a hard surface?

It reflects – it bounces back, with the angle it leaves at equal to the angle it arrived at.

D. b) Effect of superposition (two water waves meeting)

If the waves arrive in step (in phase), they add together (constructively), giving a wave with a larger amplitude than either wave alone; if out of step, they partially or fully cancel out (destructively), giving a smaller (or zero) amplitude at that point.

P2.2 Sound and energy transfer (workbook)

A. Fill in the gaps

An object must **vibrate** to cause a sound wave. As it moves backwards and forwards, it causes **vibrations** in the medium to do the same. Sound needs a medium, whether solid, liquid, or gas, to travel through. It cannot travel through a **vacuum**. Sound travels fastest in a **solid** – this is because the particles in a solid are closer together than in a liquid or a gas, so the vibration is passed along more **quickly**. **Light** travels almost a million times faster than sound and does not need a **medium** to travel through.

B. Why speed of sound differs between solids and gases

In a solid, the particles that make up the medium are packed closely together, so a vibration quickly passes from one particle to its close neighbours, letting the sound wave travel faster. In a gas, the particles are far apart, so a vibrating particle must travel further before it collides with and passes on the vibration to the next particle, making the sound wave travel more slowly through the gas medium.

C. What does supersonic travel mean?

Travelling faster than the speed of sound (faster than about 340 m/s in air), so the aircraft outruns its own sound waves.

D. Firework: time to hear vs see the explosion

The flash of light reaches the observer's eyes almost instantly (light travels at 300 000 000 m/s), but the bang travels far more slowly (340 m/s in air), so it takes noticeably longer to arrive – the flash is always seen before the bang is heard, with the delay growing the further away the observer is.

E. Why we will never hear the Sun

Sound needs a medium (particles) to travel through and cannot travel through a vacuum. The space between the Sun and Earth is (almost entirely) a vacuum, so any sound the Sun produces has no medium to travel through and can never reach Earth – whereas light can travel through a vacuum, which is why we can still see the Sun.

P2.3 Loudness and pitch (workbook)

A. Fill in the gaps

A loud sound has a bigger **amplitude** than a quieter sound. A sound with high pitch has a higher **frequency**, which is measured in **hertz** (Hz). Humans cannot hear sound waves with a frequency below **20** Hz; these are called **infrasound**. Humans cannot hear sound waves with a frequency above 20 000 Hz; these are called **ultrasound**. The range of frequencies that a human or animal can hear is called the **audible** range. Many animals can hear frequencies that are much **higher** than the frequencies we can hear.

B. Human vs other animals' hearing range

Humans typically hear 20–20 000 Hz. Many animals (e.g. bats, dolphins, dogs) hear a wider range extending to much higher frequencies, letting them hear ultrasound that's inaudible to humans.

C. a)/b) Sketching louder / lower-pitched waves

a) A louder wave: same wavelength/frequency as the original, but with taller peaks and deeper troughs (bigger amplitude). b) A lower-pitched wave: same amplitude as the original, but with fewer, more widely-spaced peaks and troughs (a longer wavelength, lower frequency).

D. What would each species hear for a 1000 Hz sound wave?

Human (20–20 000 Hz): **Yes**, hears it – 1000 Hz is within range.
Bat (2000–110 000 Hz): **No**, cannot hear it – 1000 Hz is below the bat's lower limit of 2000 Hz.
Dog (67–45 000 Hz): **Yes**, hears it – 1000 Hz is within range.

P2.4 Detecting sound (workbook)

A. Fill in the gaps

Your **ear** detects sound waves by the outer ear or **pinna** collecting the waves and directing them along the **auditory** canal to the middle ear where the **eardrum** vibrates. These vibrations pass to the **ossicles**, tiny bones that **amplify** the sound and pass the vibration through the oval window to the inner ear or **cochlea**. Here the liquid and **hair** cells vibrate. Electrical signals are transmitted from here via the auditory **nerve** to the **brain** and you hear the sound. Sound intensity is measured in **decibels** (dB). Exposure to very loud sounds can permanently **damage** the hair cells in the cochlea so that you can become deaf. A microphone has a flexible plate called a **diaphragm** that vibrates, producing an **electrical** signal. Loudspeakers vibrate, converting the electrical signal back to sound via an **amplifier** to make the sound louder.

B. How ossicles and the auditory nerve help you hear

a). Ossicles: tiny bones in the middle ear that amplify the vibration from the eardrum and pass it on (via the oval window) to the cochlea.

b). Auditory nerve: carries the electrical signal produced in the cochlea to the brain, where it's interpreted as sound.

C. How the pinna, eardrum and cochlea transfer vibrations

a). Pinna: collects/funnels sound waves and directs them into the auditory canal.

b). Eardrum: vibrates when sound waves reach it, passing the vibration on to the ossicles.

c). Cochlea: contains liquid and hair cells; the liquid vibrates (moved via the oval window by the ossicles), making the hair cells move and produce an electrical signal.

D. Note on hearing damage for construction workers

1. Loudness of the noise: very loud equipment can damage the hair cells in your cochlea – wear ear defenders/earplugs when operating loud machinery. 2. Duration of exposure: longer exposure raises the risk of permanent damage – take regular breaks away from noisy areas and limit continuous exposure time.

E. Similarity/difference: ear vs microphone

Similarity: both have a part that vibrates in response to sound (eardrum / diaphragm) and convert this into another kind of signal. Difference: the ear converts it into a nerve signal sent to and interpreted by the brain; a microphone converts it into an electrical signal sent to external equipment (e.g. an amplifier or recorder), with no interpretation involved.

P2.5 Echoes and ultrasound (workbook)

A. Fill in the gaps

Ultrasound is sound waves with a frequency too high for humans to hear, above **20 000** Hz. It can be used for things like **sonar**, where a beam of ultrasound is sent out by a **transmitter** under a ship. It travels through the water and **reflects** off the seabed. A **receiver** detects the reflection and measures the **time** it took to transmit and receive the reflection from the seabed. That allows us to work out how deep the sea must be. The reflection of a sound is called an **echo**. If lots of **echoes** join together to produce a longer sound, this is called a **reverberation**.

B a). Ultrasound is sound with a frequency above the upper limit of human hearing (above 20 000 Hz) – too high-pitched for a human to hear.

B b). The machine sends ultrasound pulses into the body; they reflect (echo) off boundaries between different tissues, including the fetus's surface. A receiver detects the returning echoes, and the time each takes to return (with the known speed of sound in tissue) gives the distance to each surface, which is processed into an image.

B c). Sonar (on ships): a transmitter sends ultrasound into the water, which reflects off the seabed or objects; a receiver detects the echo, and the time delay gives the depth or the objects' position.

C. Bats use echolocation to find food/navigate in the dark: they emit high-frequency ultrasound pulses, which reflect off nearby objects (e.g. flying insects) and return as echoes; by timing and directionally detecting these echoes, a bat can work out an object's distance, direction, size and movement without needing to see it.

P1 Chapter 2 Pinchpoint

Pinchpoint question

The scan of the four oscilloscope traces (main wave, X, Y, Z) is too low-resolution to count wave cycles with full confidence, so treat this as best-effort:

Based on the trace shapes described, X appears to show fewer cycles than the reference wave (a lower frequency, so a *lower* pitch, not higher), while Y and Z both show more cycles across the same width (a higher frequency, so a higher pitch). Correct answer: **C – Y and Z**.

Follow-up activity A

a). R is louder than the reference wave (R has the larger amplitude of the two).

b). Q has the higher pitch (Q shows more wave cycles across the same width, i.e. a higher frequency, even though it is quieter than R).

Follow-up activity B

Wave S has more wave cycles packed into the same width than wave T – its peaks are closer together (a shorter wavelength, i.e. a higher frequency), which is what shows S has the higher pitch.

Follow-up activity C (matches the correct Pinchpoint answer, C)

Violet light has a higher frequency than red light, so the sketched wave should have more cycles packed into the same width than wave U. Being dim (rather than bright) means a smaller amplitude, so the peaks and troughs should also be drawn shorter than wave U's.

Follow-up activity D

a). On both V and W: amplitude = the vertical distance from the centre line up to a peak; wavelength = the horizontal distance between two equivalent points on consecutive waves (e.g. peak to peak).

b). **Half as big as wave V's wavelength.** Frequency and wavelength are inversely related: since wave W fits twice as many waves into the same space as wave V (a higher frequency), its wavelength must be half as long.

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